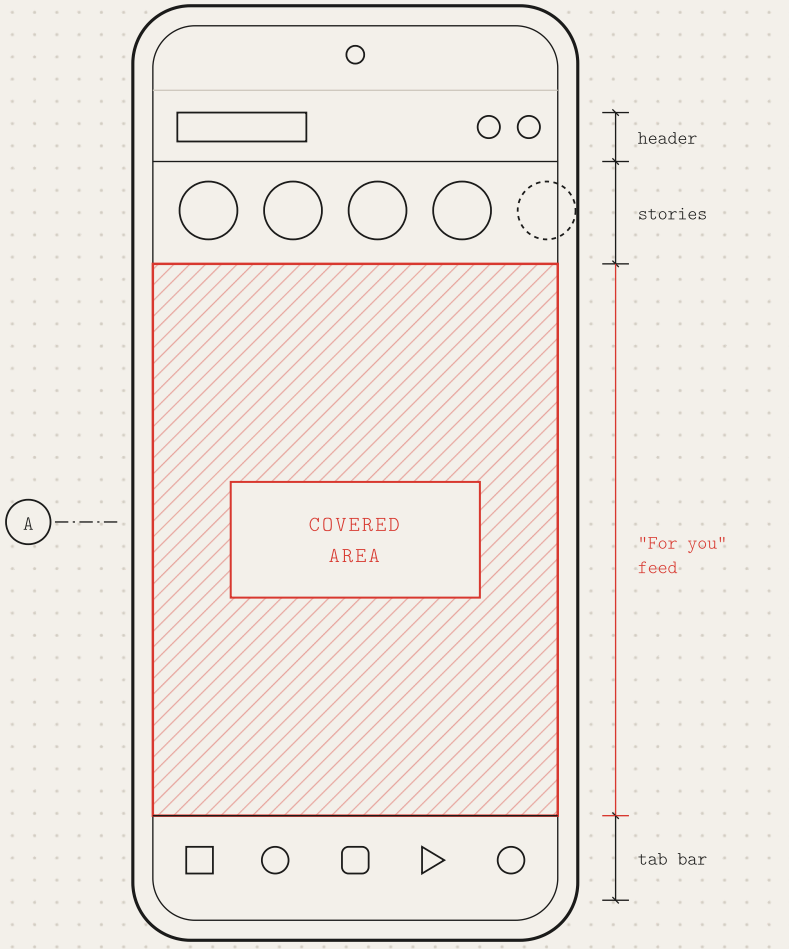
PRODUCT ENGINEERING · ANDROID ACCESSIBILITY

Sin Shorts.

Selective short-video blocking for Android: removes Shorts, Reels and Spotlight without blocking the apps that contain them.



ELEVATION - INSTAGRAM, "FOLLOWING ONLY" MODE

0	77 KB	0
NETWORK PERMISSIONS	APK SIZE	DEPENDENCIES

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01 - PROJECT BRIEF

Block the section, not the app

Typical blockers work per app: either you use YouTube or you don't. Short-form video lives inside apps that are useful for other reasons.

Goal. Remove short-form video and infinite scroll (Shorts, Reels, Spotlight, Instagram's suggested feed and Explore) while keeping the useful features intact: long-form video, messages, stories and search.

Approach. An Android accessibility service that receives screen changes, recognises each app's view hierarchy through declarative rules and responds with the minimal action: pressing Back, or placing an opaque layer over one specific region.

Design constraints. No network access, no third-party libraries, no server, and user-editable configuration so it survives app redesigns without shipping a new release.

TECHNICAL SHEET

Platform	Android 8.0 – 16 · minSdk 26 · targetSdk 35
Language	Java 8 · UI built in code, no XML layouts, no AppCompat
Core API	AccessibilityService · AccessibilityNodeInfo · GestureDescription · TYPE_ACCESSIBILITY_OVERLAY
Build	Minimal toolchain without Gradle: aapt · javac · dx · zipalign · apksigner (v2/v3 signing)
Distribution	Signed APK · open source, MIT licence · CI on GitHub Actions
Covered apps	YouTube · Instagram · Facebook · Snapchat · TikTok · Chrome, Firefox, Brave, Edge, DuckDuckGo

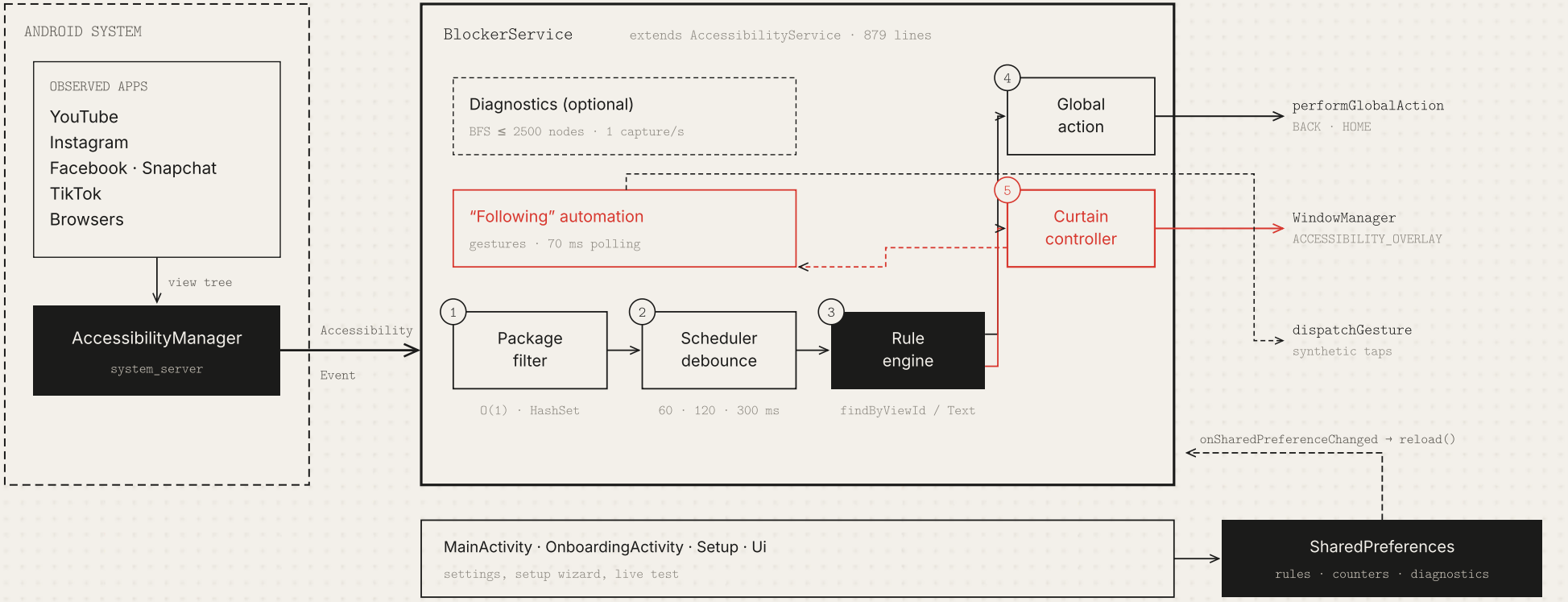
2.4k	8	11
LINES OF JAVA	CLASSES	ITERATED RELEASES
19	70ms	1
BUILT-IN RULES	MIN. POLLING CYCLE	DECLARED PERMISSION

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02 - GENERAL PLAN

System architecture

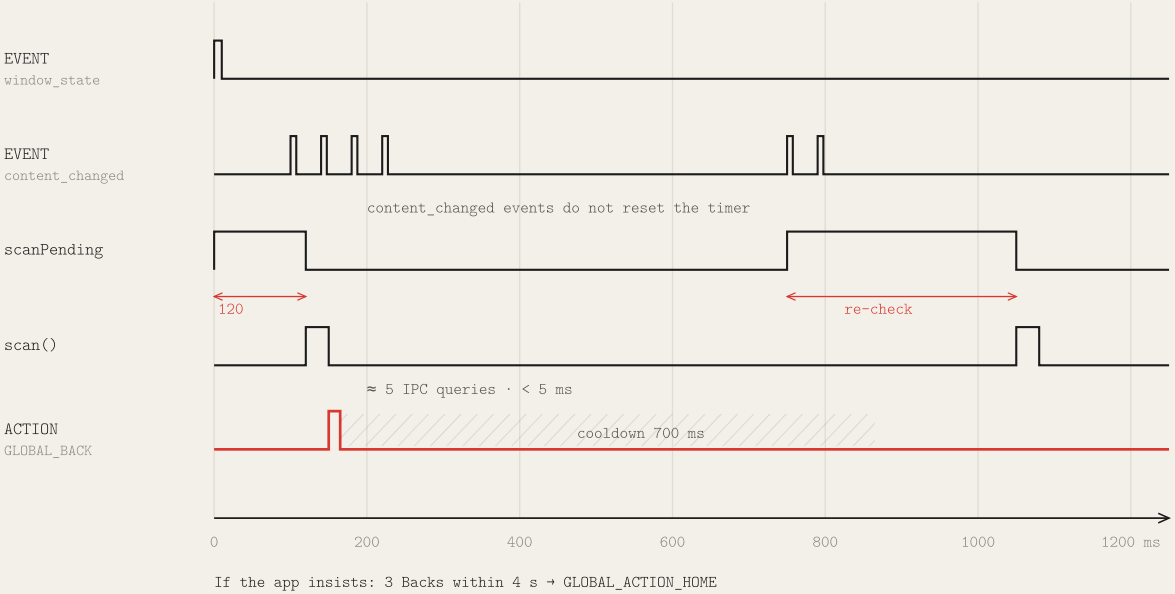
A single process. The accessibility service is the core; screens only write configuration to `SharedPreferences` and the service hot-reloads it through a listener.



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03 - TIMING

From event to action



If the app insists: 3 Backs within 4 s → GLOBAL_ACTION_HOME

PRINCIPLE

There is no periodic polling. The system reports every screen change and the service decides whether it deserves a check. The first filter discards any app not on the list in constant time.

The scheduler coalesces bursts: a window change is checked after 60–120 ms; content changes at most once every 300 ms.

PARAMETERS

CONSTANT	VALUE	PURPOSE
SCAN_DELAY_STATE	60 / 120 ms	Instagram / others
SCAN_DELAY_CONTENT	300 ms	Rate ceiling
ACTION_COOLDOWN	700 ms	Prevents double Back
RETRY_WINDOW	4 s · 3	Escalation to HOME
settleRuns	250 + 2×500 ms	Re-check after animations
SOFT_HIDE_CONFIRM	300 ms	Hysteresis on exit

> Estimated cost during video playback: on the order of 1 % of the observed app's own consumption, which is dominated by the screen and decoding.

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04 - SPECIFICATION

A one-line rule language

Every blocking behaviour is data, not code. Built-in and user rules share one format. When an app changes its design, the fix is a single line typed on the phone, with no rebuild.

GRAMMAR

```
rule     := packages | type
packages := id ( , id )*
type     := app | key : value
key      := id | idsel | text | textsel | url
```

TYPES

`id`	Visible view with that resource ID
`idsel`	Same, and selected (active tab, ±2 levels)
`text`	Text or description contains the value
`textsel`	Same, and selected
`url`	Address bar contains the value
`app`	Whole app (sends to the home screen)

BUILT-IN RULES (EXCERPT)

```
# YouTube Shorts
com.google.android.youtube | id:reel_recycler
com.google.android.youtube | id:reel_watch_player
com.google.android.youtube | id:shorts_player
# Instagram Reels
com.instagram.android | idsel:clips_tab
com.instagram.android | id:clips_viewer_view_pager
# Facebook · Snapchat
com.facebook.katana | textsel:reels
com.snapchat.android | textsel:spotlight
# Browsers
com.android.chrome,org.mozilla.firefox,… | url:youtube.com/shorts
# TikTok (off by default)
com.zhiliaoapp.musically | app
```

BY ID

Language-independent and very cheap: a single system query (`findAccessibilityNodeInfosByViewId`). The preferred strategy.

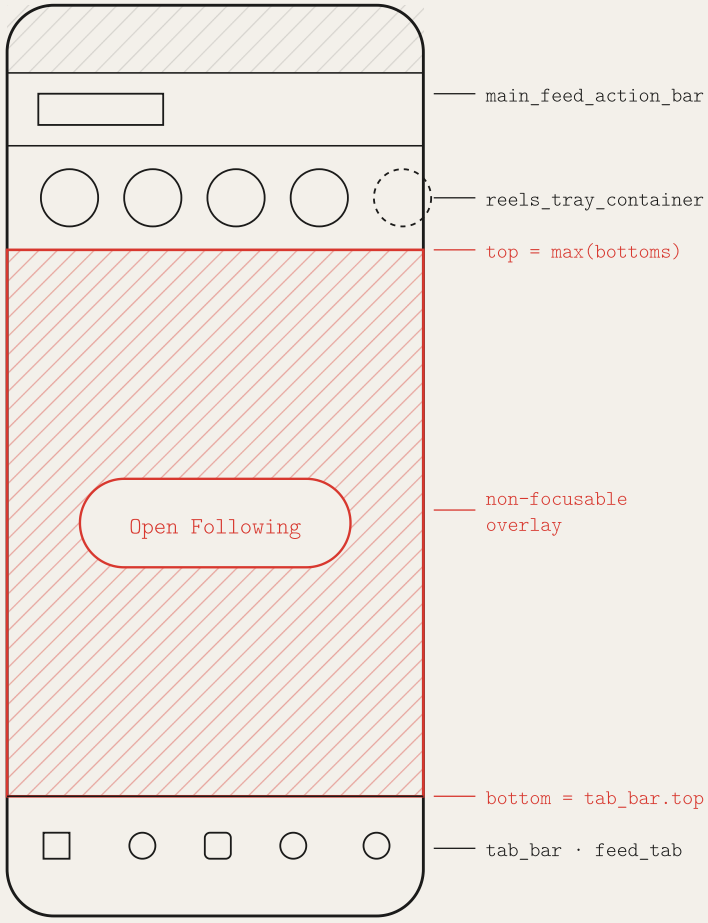
BY TEXT

For apps that obfuscate their IDs (Facebook, Snapchat). Language-dependent, so it is combined with the “selected” state.

> **Diagnostic mode.** Walks the view tree breadth-first (≤ 2500 nodes) and lists the visible IDs and selected elements of any app, so a new rule can be written without developer tools.

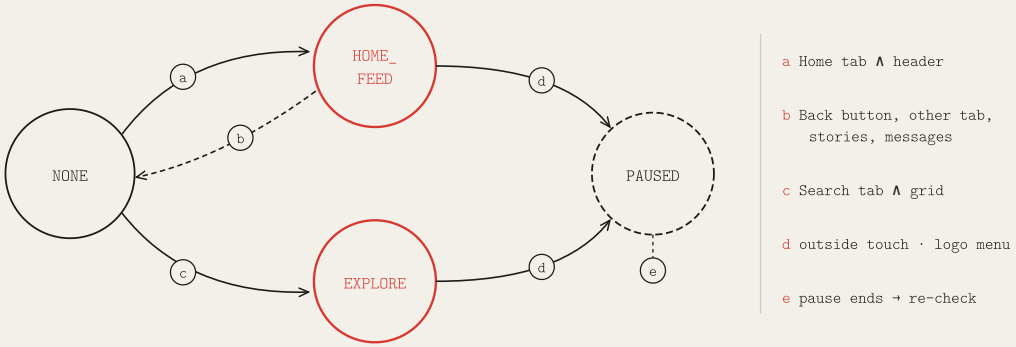
05 - SECTION A-A

The curtain



Instagram's home mixes in suggested posts without end. Instead of blocking the app, only the feed is covered with a `TYPE_ACCESSIBILITY_OVERLAY` window, with no extra permissions. Its geometry is recomputed on every check from the actual bounds of the header, the stories tray and the tab bar. Messages, stories and notifications stay reachable.

STATE MACHINE



WINDOW

- `FLAG_NOT_FOCUSABLE`: keyboard and Back still go to Instagram
- `FLAG_NOT_TOUCH_MODAL`: only absorbs touches inside its area
- `FLAG_WATCH_OUTSIDE_TOUCH`: withdraws on the very *touch-down* outside it

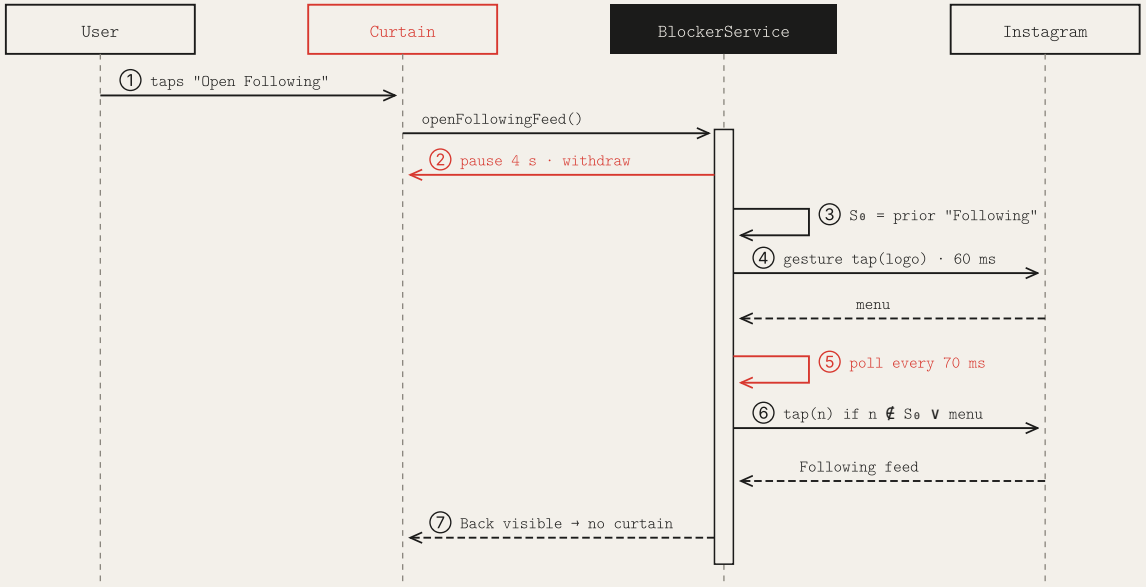
STABILITY

- Story viewer detected by ID: never covered
- Three re-checks after every change, in case an animation ends silently
- 300 ms hysteresis for ambiguous readings during transitions

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06 - SEQUENCE

“Open Following” in one tap



no fixed wait: reacts as soon as the menu exists

THE PROBLEM

Instagram exposes no direct link to the chronological feed. You have to tap the logo and pick “Following” from a drop-down menu, and the same label can also appear elsewhere on screen.

THE SOLUTION

- **Locate the logo** by name, by ID or, failing that, as the widest element in the header
- **Real taps** via `dispatchGesture`: some views ignore `ACTION_CLICK`
- **State diff.** Only a “Following” that did not exist before the tap (S_o) is accepted, or the full menu (“Following” and “Favourites” together)
- **Failure log.** On failure, the relevant view tree is saved for cable-free debugging

> Blind polling with a fixed wait was reliable but slow (≈ 0.45 s). Aggressive polling without S_o tapped the wrong elements. The state diff allows fast polling with no false positives.

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07 - PRIVACY BY DESIGN

No network, no doubts

An accessibility service can read any screen. Trust is not requested, it is guaranteed by construction: the manifest does not declare `INTERNET`, so the operating system prevents any data from leaving the device.

PERMISSION SURFACE

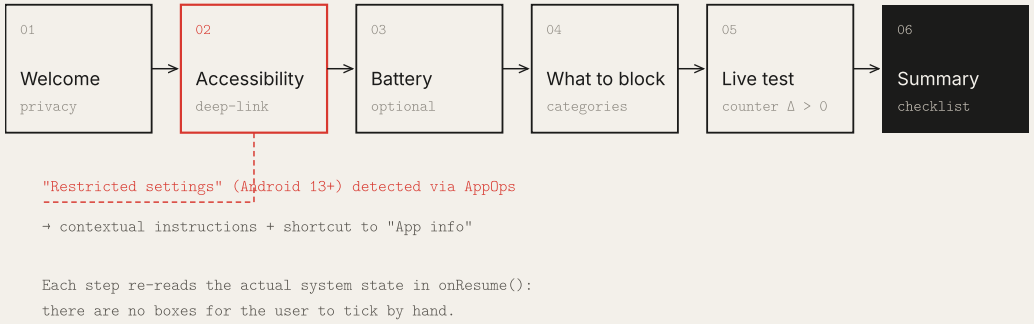
ITEM	USE
BIND_ACCESSIBILITY_SERVICE	Receive events and read the view tree
canPerformGestures	Synthetic taps for the “Following” button
IGNORE_BATTERY_OPT.	Optional · keeps the system from stopping the service
INTERNET	Not declared

DATA AT REST

Settings, rules, daily counters and, only when enabled, diagnostic captures (view IDs, never content). Everything lives in the app’s private storage.

SETUP WIZARD

The app detects the state of each system setting on its own when the user returns from Settings, and advances by itself. The last step is a live test that opens a real Short and verifies the block through the internal counter.



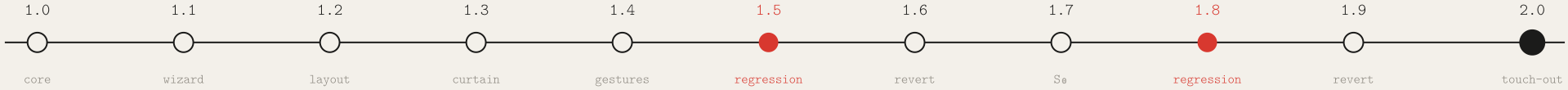
Distribution. Play Protect blocks direct installation of accessibility apps downloaded from the web. Publishing on Google Play requires an explicit consent screen and an API-use declaration.

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08 - ITERATION LOG

Measure on device, fix the root cause

11 releases, each validated on a Pixel running a recent Android. Two regressions were reverted rather than patched.



RELEASE	OBSERVED SYMPTOM	ROOT CAUSE	FIX
1.2	The wizard rendered empty, with its buttons mid-screen	Spacer with `wrap_content` and a weight: an empty `View` claims all available height	Fixed 1 px height; the content gets its space back
1.4	“Open Following” did nothing	The curtain itself covered the menu; `ACTION_CLICK` ignored by the logo view	Withdraw the layer before acting and use `dispatchGesture` (compiled against API 33)
1.5	Stories opened instead of “Following”	Immediate polling tapped a “Following” label outside the menu	Reverted in 1.6; solved in 1.7 with the S_o state diff
1.7	Curtain “stuck” when switching screens	Instagram emits no event when some animations finish	Three scheduled re-checks after every change
1.8	The curtain lingered 2 s over stories	Hysteresis also applied to a case that had to be immediate	Reverted in 1.9; hysteresis limited to leaving the app
2.0	Delay when tapping a story	Instagram emits no `VIEW_CLICKED` in the stories tray	`FLAG_WATCH_OUTSIDE_TOUCH`: withdrawal on *touch-down* + story-viewer detection by ID

LESSON 01

Revert a regression, don’t patch it. Returning to the last good state and reintroducing the change once its cause was understood was faster than stacking patches.

LESSON 02

Instrument before guessing. The failure log stores, on the phone itself, the view tree at the exact moment of the error: debugging without a cable or ADB.

LESSON 03

Perceived latency rules. 300 ms that are correct but visible feel like a bug. Reacting on *touch-down* changed the experience more than optimising the polling did.

Author’s role: product and requirements definition, real-device testing, failure diagnosis and validation of every iteration. Implementation developed with AI assistance (Claude, Anthropic).

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09 - LIMITS

What it cannot do

- **iOS.** The system sandboxes every app and offers no equivalent API. The only option there is a Safari extension for the web versions.
- **Dependence on third-party UI.** View IDs change with updates. Editable rules and the diagnostic mode mitigate this.
- **Apps with obfuscated IDs.** Facebook and Snapchat force text matching, which depends on the language.
- **Distribution.** Direct installation runs into Play Protect and Android 13+ restricted settings.

NEXT STEPS

Roadmap

01 Restrict the service by package in the manifest, so the system stops notifying it about other apps

02 On YouTube, ignore `content_changed` during long-form playback

03 Consent screen and Google Play release

04 Rules updatable from a signed file, without opening the app to the network

05 Instrumented tests against recorded view trees

SKILLS

Android · Java | Accessibility API | Event-driven systems | DSL design | Timing analysis | Privacy by design | On-device debugging | Build toolchain | Setup UX

A soft real-time, event-driven system that coexists with third-party software it does not control.

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